Core Equations: Ball Mill Ore Fracture

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1 Simulation Parameters

1.1 Swept Input Parameters

Five parameters are varied across simulation runs. The default value is used whenever a parameter is held fixed.

Symbol	Description	Default	Range	Grid points
v	Impact velocity (m/s)	2.0	0.5 – 5.0	46 (step 0.1)
α	Impact angle (°)	0	−90 – 90	46 (step 4)
b	Impact parameter (m)	0	0 – R+r	45 (44 intervals)
$\chi = R/r$	Size ratio	4	2 – 20	46 (step 0.4)
μ	Friction coefficient	0.47	0 – 1	41 (step 0.025)

Table 1: Swept parameters. $b_{\max} = R + r = R(1 + 1/\chi)$ varies with χ.

1.2 Fixed Geometric Constants

Symbol	Description	Value
R	Grinding ball radius	0.025 m
g	Launch gap (mover to ore)	0.010 m
Δt	LAMMPS timestep	1×10^{-7} s

Table 2: Fixed geometric and simulation constants.

1.3 Material Properties

Property	Symbol	Limestone ore	Steel ball
Density ($\mathrm{kg\,m^{-3}}$)	ρ	2600	7800
Young's modulus (GPa)	E	77.82	208
Poisson ratio	ν	0.28	0.27
Restitution coefficient	e	0.50	0.50

Table 3: Material properties used in LAMMPS contact model.

Criterion	Symbol	Description	Value
Christensen	σ_T	Tensile strength of limestone	22.4 MPa
	σ_C	Compressive strength of limestone	147 MPa
Weibull	w	Weibull modulus	2.73
	d_0	Reference particle diameter	120 mm
	$\sigma_{pc,0}$	Crush stress at d_0	5.579 MPa

Table 4: Constants for the two fracture criteria (Sections 2 and 4).

1.4 Fracture Criterion Constants

2 Theory 1: DEM + Christensen Criterion

2.1 Christensen Failure Criterion

For a brittle material with uniaxial tensile strength σ_T and compressive strength σ_C, the utilisation index is:

$$U = \frac{1}{k^2}\left[\frac{\alpha_c\, k}{\sqrt{3}}\,\sigma_{kk} \;+\; \frac{1+\alpha_c}{2}\,s_{ij}s_{ij}\right] \tag{1}$$

where the material parameters are:

$$\alpha_c = \frac{\sigma_T}{\sigma_C} - 1, \qquad k = \frac{\sigma_C}{\sqrt{3}} \tag{2}$$

and the stress invariants are:

$$\sigma_{kk} = \sigma_{xx} + \sigma_{yy} + \sigma_{zz} \quad \text{(hydrostatic trace)} \tag{3}$$

$$s_{ij} = \sigma_{ij} - \tfrac{1}{3}\sigma_{kk}\,\delta_{ij} \quad \text{(deviatoric stress tensor)} \tag{4}$$

$$s_{ij}s_{ij} = s_{xx}^2 + s_{yy}^2 + s_{zz}^2 + 2\big(s_{xy}^2 + s_{xz}^2 + s_{yz}^2\big) \tag{5}$$

Fracture criterion: $U \geq 1$.

2.2 DEM Implementation Pipeline

LAMMPS outputs a per-atom virial stress tensor W_{ij} via `compute stress/atom`. The following steps convert it to U:

1. **Identify the first contact window.** From the position dump (`op1.dump`), find all timesteps where the mover–ore overlap $\delta = (R + r) - |\mathbf{x}_{\text{mover}} - \mathbf{x}_{\text{ore}}| > 0$. Only the first continuous contact event is analysed.

2. **Cauchy stress.** Convert the LAMMPS virial to a volumetric Cauchy stress:

$$\sigma_{ij} = \frac{W_{ij}}{V}, \qquad V = \frac{\pi}{6}\,d^3 \tag{6}$$

   where d is the ore diameter read from `op1.dump`.

3. **Branch-length correction.** The virial sums over the full centre-to-centre branch vector between contacting atoms; only the fraction lying *inside* the ore should contribute. The corrected stress is:

$$\sigma_{ij}^{\text{corr}} = \lambda\,\sigma_{ij}, \qquad \lambda = \frac{2}{1+\chi} \tag{7}$$

   At $\chi = 4$: $\lambda = 0.4$.

4. **Evaluate U.** Apply Eq. (1) to $\sigma_{ij}^{\text{corr}}$ at each timestep in the window and record $U_{\max}$.

5. **Fracture decision.** Fracture is predicted if $U_{\max} \geq 1$.

3 Theory 2: Analytic Energy Fracture Criterion

3.1 Assumptions

1. Uniaxial collision (head-on, $b = 0$).
2. No damping — kinetic energy is fully converted to elastic strain energy.

3.2 Critical Force and Deflection

The maximum contact stress under a Hertzian load F acting on the ore (radius r) is:

$$\sigma_{xx} = \frac{3}{2}\frac{F}{\pi r^2} \tag{8}$$

Fracture occurs at $\sigma_{xx} = \sigma_c$, giving the critical force and deflection:

$$F_c = \frac{2}{3}\pi r^2 \sigma_c \tag{9}$$

$$\delta_c = \left(\frac{3\,F_c}{4\,E_{\text{eff}}\,R_{\text{eff}}^{1/2}}\right)^{2/3} \tag{10}$$

where the effective radius combines the mover (R) and ore (r):

$$\frac{1}{R_{\text{eff}}} = \frac{1}{r} + \frac{1}{R} \;\Rightarrow\; R_{\text{eff}} = \frac{\chi\, r}{1+\chi} \tag{11}$$

and the effective Young's modulus combines both materials:

$$\frac{1}{E_{\text{eff}}} = \frac{1-\nu_1^2}{E_1} + \frac{1-\nu_2^2}{E_2} \tag{12}$$

3.3 Critical Strain Energy

The elastic energy stored in *one* Hertz contact at the critical deflection is:

$$U_{\text{pair}} = \int_0^{\delta_c} F\,d\delta = \frac{4}{3}E_{\text{eff}}R_{\text{eff}}^{1/2}\int_0^{\delta_c}\delta^{3/2}\,d\delta = \frac{8}{15}\,E_{\text{eff}}\,R_{\text{eff}}^{1/2}\,\delta_c^{5/2} \tag{13}$$

The ore is compressed between *two* balls (mover and fixed), so the total critical energy is $E_c = 2\,U_{\text{pair}}$:

$$E_c = \frac{16}{15}\,E_{\text{eff}}^{-2/3}\,r^3\left(\frac{\chi}{1+\chi}\right)^{-1/3}\sigma_c^{5/3}\left(\frac{\pi}{2}\right)^{5/3} \tag{14}$$

3.4 Head-On Critical Velocity

Setting the mover's kinetic energy equal to E_c (mover volume $V_{\text{mover}} = \frac{4}{3}\pi R^3 = \frac{4}{3}\pi(\chi r)^3$):

$$\frac{1}{2}\rho_{\text{steel}}\cdot\frac{4}{3}\pi(\chi r)^3\cdot v_c^2 = E_c \tag{15}$$

$$\boxed{v_c = \frac{2^{2/3}\sqrt{5}\,\pi^{1/3}\,\sigma_c^{5/6}\,\chi^{-5/3}(1+\chi)^{1/6}}{5\,\rho_{\text{steel}}^{1/2}\,E_{\text{eff}}^{1/3}}} \tag{16}$$

At default parameters ($\chi = 4$, $\sigma_c = 147$ MPa, $\rho_{\text{steel}} = 7800$ kg/m^3): $v_c = 2.483$ m/s.

3.5 Extension to Oblique Impact

For a general impact, only the effective compression velocity does work on the ore:

$$v_x = |v| \cos\alpha \cos\gamma, \qquad \gamma(b, \chi) = \arcsin\left(\frac{b}{R + R/\chi}\right) \tag{17}$$

Substituting $v \to v_x$ into the head-on energy balance gives the general critical velocity:

$$\boxed{v_c(\alpha,\, b,\, \chi) = \frac{2^{2/3}\sqrt{5}\,\pi^{1/3}\,\sigma_c^{5/6}\,\chi^{-5/3}(1+\chi)^{1/6}}{5\,\rho_{\text{steel}}^{1/2}\,E_{\text{eff}}^{1/3}\,|\cos\alpha|\sqrt{1 - \left(\dfrac{b}{R + R/\chi}\right)^2}}} \tag{18}$$

4 Theory 3: DEM + Weibull Weakest-Link Criterion

4.1 Weibull Size-Scaling Law

The characteristic particle crushing stress scales with particle diameter d_p according to the Weibull weakest-link model:

$$\sigma_{pc}(d_p) = \sigma_{pc,0}\left(\frac{d_p}{d_0}\right)^{-3/w} \tag{19}$$

The corresponding failure load (using $F = \sigma_{pc}\, d_p^2$) is:

$$F_{f,W}(d_p) = \sigma_{pc,0}\left(\frac{d_p}{d_0}\right)^{-3/w} d_p^2 \tag{20}$$

where $d_p = 2r = 2R/\chi$ is the ore diameter, and w, d_0, $\sigma_{pc,0}$ are the Weibull constants from Table 4.

4.2 Compression Axis

The ore is loaded simultaneously by the mover and the fixed ball. Let $\hat{\mathbf{r}}_{21}$ and $\hat{\mathbf{r}}_{23}$ be the unit vectors from the ore centre toward the fixed ball and mover, respectively:

$$\hat{\mathbf{r}}_{21} = \frac{\mathbf{x}_{\text{fixed}} - \mathbf{x}_{\text{ore}}}{|\mathbf{x}_{\text{fixed}} - \mathbf{x}_{\text{ore}}|}, \qquad \hat{\mathbf{r}}_{23} = \frac{\mathbf{x}_{\text{mover}} - \mathbf{x}_{\text{ore}}}{|\mathbf{x}_{\text{mover}} - \mathbf{x}_{\text{ore}}|} \tag{21}$$

The compression axis is the bisector of these two directions:

$$\hat{\mathbf{n}} = \frac{\hat{\mathbf{r}}_{21} - \hat{\mathbf{r}}_{23}}{|\hat{\mathbf{r}}_{21} - \hat{\mathbf{r}}_{23}|} \tag{22}$$

This sign convention ensures that both contact force projections are positive during compression (verified at head-on: $\hat{\mathbf{n}} = -\hat{x}$, $\mathbf{F}_{\text{mover}} \cdot \hat{\mathbf{n}} > 0$, $-\mathbf{F}_{\text{fixed}} \cdot \hat{\mathbf{n}} > 0$).

4.3 Effective Compression Force

Let $\mathbf{F}_{1\leftarrow 2}$ and $\mathbf{F}_{1\leftarrow 3}$ be the forces on the ore from the mover and fixed ball, respectively (recovered from `forces.dump` as described in Section 7). Their projections onto $\hat{\mathbf{n}}$ are:

$$F_2^n = \mathbf{F}_{1\leftarrow 2} \cdot \hat{\mathbf{n}}, \qquad F_3^n = -\mathbf{F}_{1\leftarrow 3} \cdot \hat{\mathbf{n}} \tag{23}$$

The effective compression force is the average of the two (a symmetric diametral loading is assumed, consistent with the Weibull test geometry):

$$F_{\text{eff}} = \max\left(\frac{F_2^n + F_3^n}{2},\ 0\right) \tag{24}$$

4.4 Weibull Utilisation Index and Fracture Criterion

$$U_W = \frac{F_{\text{eff}}}{F_{f,W}(d_p)} \tag{25}$$

Fracture criterion: $U_W \geq 1$.

4.5 DEM Implementation Pipeline

1. **Identify the first contact window** from `op1.dump` (same procedure as Theory 1).
2. **Recover contact forces.** Read per-atom forces on ore (atom 1) and mover (atom 2) from `forces.dump` and apply Newton's third law (Eq. 7) to obtain $\mathbf{F}_{1\leftarrow 2}$ and $\mathbf{F}_{1\leftarrow 3}$ at each timestep.
3. **Compute $\hat{\mathbf{n}}$** from atom positions in `op1.dump` at each timestep.
4. **Evaluate** U_W at each timestep in the window; record $U_{W,\max}$.
5. **Fracture decision:** fracture predicted if $U_{W,\max} \geq 1$.

5 Theory 4: Analytic Weibull Energy Criterion

5.1 Specific Fracture Energy

Integrating the Hertz contact law $F = k\delta^{3/2}$ over two identical contacts up to the failure load, normalising by the particle volume $V_p = \frac{4}{3}\pi R_S^3$, and substituting the Weibull size-scaling law (Eq. 22 of the companion derivation) gives the size-dependent specific fracture energy:

$$E_{pc}(d_p) = \frac{4.989}{\pi}\,(E^*)^{-2/3}\left(\frac{R_S}{R^*}\right)^{1/3}\sigma_{pc,0}^{5/3}\,d_0^{5/w}\,d_p^{-5/w} \tag{26}$$

where the Hertz contact quantities are:

$$\frac{1}{E^*} = \frac{1-\nu_S^2}{E_S} + \frac{1-\nu_B^2}{E_B}, \qquad \frac{1}{R^*} = \frac{1}{R_S} + \frac{1}{R_B}, \qquad R_S = \frac{d_p}{2} = \frac{R}{\chi} \tag{27}$$

The ratio $R_S/R^* = (1+1/\chi)^{-1}\cdot(1+\chi)/\chi$ simplifies to $(1+1/\chi)$ when $R_B \gg R_S$ (plate limit), and the $4.989/\pi$ prefactor reduces to the Zhang et al. platen-test result in that limit.

5.2 Energy Balance and Critical Velocity

The available kinetic energy along the compression axis is (using the effective compression velocity $v_x = v|\cos\alpha|\cos\gamma$ from Section 3, with $\gamma(b,\chi) = \arcsin(b/(R+R/\chi))$):

$$W_{\text{in}} = \tfrac{1}{2}\,M_{\text{proj}}\,v_x^2 = \tfrac{1}{2}\,M_{\text{proj}}\,(v\cos\alpha\cos\gamma)^2, \qquad M_{\text{proj}} = \tfrac{4}{3}\pi R_B^3\,\rho_B \tag{28}$$

Setting $W_{\text{in}} = V_p\,E_{pc}(d_p)$ and solving for the critical velocity:

$$\boxed{v_c(\alpha,b,\chi) = \frac{1}{|\cos\alpha|\,\cos\gamma(b,\chi)}\sqrt{\frac{2\,V_p\,E_{pc}(d_p)}{M_{\text{proj}}}}} \tag{29}$$

Substituting $V_p/M_{\text{proj}} = \chi^{-3}/\rho_B$, $\cos\gamma = \sqrt{1-(b/(R+R/\chi))^2}$, and the full expression for E_{pc} gives explicitly:

$$v_c(\alpha,b,\chi) = \frac{1}{|\cos\alpha|\sqrt{1-\left(\dfrac{b}{R+R/\chi}\right)^2}}\sqrt{\frac{2}{\rho_B\,\chi^3}\cdot\frac{4.989}{\pi}\,(E^*)^{-2/3}\left(\frac{R_S}{R^*}\right)^{1/3}\sigma_{pc,0}^{5/3}\,d_0^{5/w}\,d_p^{-5/w}} \tag{30}$$

Fracture criterion: fracture predicted when $v \geq v_c(\alpha,b,\chi)$.

6 Geometry-Limiting Regime (Trapping Condition)

The four theories above govern fracture *once the ore is trapped* between the two balls. Whether trapping occurs at all is a separate question determined purely by collision geometry and friction — independent of velocity and fracture model.

6.1 Half-Angle of Approach

Using $\gamma(b, \chi)$ from Eq. (18), define θ as the half-angle between the mover–ore contact vector $\hat{\mathbf{r}}_{23}$ and the x-axis:

$$\theta(\alpha, b, \chi) = \tfrac{1}{2}(\alpha + \gamma(b, \chi)) \tag{31}$$

6.2 Force Balance and Coulomb Cone

At each contact the ore feels a normal force N_i and a tangential (friction) force T_i:

$$\mathbf{F}_i = N_i(-\hat{\mathbf{r}}_i) + T_i \hat{\boldsymbol{\tau}}_i \tag{32}$$

Coulomb's law requires $|T_i| \leq \mu N_i$, so the admissible resultant $\mathbf{F}_i$ lies inside a cone of half-angle $\phi = \arctan \mu$ about the inward normal $-\hat{\mathbf{r}}_i$. For the ore to be *trapped* (not ejected sideways), the two contact forces must cancel in the transverse direction, which requires $\phi \geq \theta$. The trapping boundary is:

$$\boxed{\mu = \tan\theta = \tan\left[\tfrac{1}{2}(\alpha + \gamma)\right]} \tag{33}$$

6.3 Critical Angle and Critical Offset

Substituting $\theta = \frac{1}{2}(\alpha + \gamma)$ and rearranging gives the critical impact angle (ore escapes if $|\alpha| > \alpha_c$):

$$\boxed{\alpha_c(\mu, b, \chi) = 2\arctan(\mu) - \arcsin\left(\frac{b}{R + R/\chi}\right)} \tag{34}$$

Rearranging for b gives the critical offset (ore escapes if $b > b_c$):

$$\boxed{b_c(\alpha, \mu, \chi) = \left(R + \frac{R}{\chi}\right)\sin(2\arctan\mu - \alpha)} \tag{35}$$

At default parameters ($\chi = 4$, $\mu = 0.47$, $b = 0$): $\alpha_c = \pm\, 2\arctan(0.47) \approx \pm 50.75°$.

7 Simulation Setup

7.1 Three-Sphere Geometry

Three spheres are placed along the x-axis:

- **Atom 3** — fixed ball (anvil), radius R, centre at the origin.
- **Atom 1** — ore, radius $r = R/\chi$, placed tangent to atom 3 at $\mathbf{x}_{\text{ore}} = (R + r,\, 0,\, 0)$.
- **Atom 2** — mover (projectile), radius R, launched toward the ore at speed v, angle α, and transverse offset b.

The mover's initial centre and velocity components are:

$$D = R + r + g \quad (g = 0.010 \text{ m launch gap}) \tag{36}$$

$$C_{2x} = (R + r) + D\cos\alpha + b\sin\alpha \tag{37}$$

$$C_{2y} = -D\sin\alpha + b\cos\alpha \tag{38}$$

$$v_{2x} = -v\cos\alpha, \qquad v_{2y} = v\sin\alpha \tag{39}$$

7.2 Run Duration

The number of timesteps is determined by solving exactly for the first-contact time T (when the mover surface reaches the ore surface):

$$A\,T^2 + B\,T + C = 0, \quad A = |\mathbf{v}|^2, \quad B = 2\,\boldsymbol{\delta}\cdot\mathbf{v}, \quad C = |\boldsymbol{\delta}|^2 - (R+r)^2 \tag{40}$$

where $\boldsymbol{\delta} = \mathbf{C}_2 - \mathbf{x}_{\text{ore}}$. Taking the smaller root and adding a 5000-step buffer for the contact event:

$$T = \frac{-B - \sqrt{B^2 - 4AC}}{2A}, \qquad N_{\text{steps}} = \left\lfloor \frac{T}{\Delta t} \right\rfloor + 5000 \tag{41}$$

7.3 Friction Mixing

LAMMPS computes the effective friction between two species as the geometric mean: $\mu_{\text{eff}} = \sqrt{\mu_1\,\mu_2}$. The ball–ball friction is fixed at $\mu_2 = 0.3$. To achieve a desired ore–ball friction μ, the ore friction is set to:

$$\mu_1 = \frac{\mu^2}{0.3} \tag{42}$$

7.4 LAMMPS Output Files

Each run produces three dump files, all written every 100 timesteps:

File	Type	Contents
`op1.dump`	Per-atom (custom)	`id type x y z vx vy vz omegax omegay omegaz diameter` Used for: contact window detection, atom positions, ore diameter
`stress.dump`	Per-atom (custom)	`id type x y z diameter c_astress[1..6]` Used for: Christensen U (virial stress tensor on ore)
`forces.dump`	Per-atom (custom)	`id fx fy fz` for atoms 1 (ore) and 2 (mover) Used for: Weibull U_W (individual contact forces via Newton's 3rd law)

Table 5: LAMMPS output files. All three are deleted after post-processing to save disk space.

The individual ore–ball contact forces are recovered from `forces.dump` as:

$$\mathbf{F}_{1\leftarrow 2} = -\mathbf{f}_{\text{mover}}, \qquad \mathbf{F}_{1\leftarrow 3} = \mathbf{f}_{\text{ore}} + \mathbf{f}_{\text{mover}} \tag{43}$$

which follows from Newton's third law and the fact that the mover has no wall contact during the relevant timesteps.